Section B: YOUR SOLUTION AND ITS COMPETITIVE EDGE

a. What does the project hope to achieve?
Describe how your solution can address the challenge statement

The challenge statement requests a new way to estimate of equipment and vehicles at the construction site based on available video footage data from existing video cameras on site.

The project we propose aims to provide a first-of-a-kind demonstration of how data analytics models can be used to measure and reduce in the construction industry.

The proposed solution will enable challenge statement owner, Obayashi, to extract useful information from videos, images, and other data sources. The platform already allows the user to connect live API feeds for structured datasets and upload any file format (e.g., csv, pdf, image, etc.). These data ingestion capabilities can be extended through the project to include live video streaming as well static video recordings, depending on the user's response requirements.

Although the challenge statement owner has requested to take into account only video as the input for data analysis, our current platform can accommodate additional data sources, such as structured data. The reason we propose 'data fusion' is because video analytics on its own can sometimes be inaccurate and may require a lot of computing costs to store and process, depending on the volume of footage that needs to be considered, creating an unintended consequences upstream through the additional emissions generated by computing on the premise or the cloud.

We propose to extend the data input considerations from videos to other data sources, such fuel invoices and equipment usage log files (digital or print). The exact format of the data input is flexible. Our current platform can accommodate any type of file and API, including video live streams after some modifications. Further to the data processing and analytics modelling, additional functionality can be added to provide automated reporting in a standard text format that's required by many standardized sustainability frameworks. The outputs can be delivered through the existing communication channels, such as email and pdf (available for download). This minimizes the need for primary users, such as construction supervisors, to shift to digital dashboards on tablets or smartphones, which are often ill-equipped for the environment and internet conditions found in many construction sites.

b. What is the new product / service / module / technology component that you are planning to develop if selected for BEAMP?
Describe its functions and features. Please include illustrations or mock-ups to help us visualise the product / service / module / technology component.

The new product modules we plan to develop through BEAMP can be grouped into three categories.

First, we want to enhance the data engineering pipeline to provide automated ingestion and verification of source files that the user already has – without needing to adopt new enterprise systems or complicated machinery. For instance, the platform already allows the user to upload any type of source file or connect live data via an API. We aim to extend this capability to videos for both historical files and real-time live streams.

After completing the ASD track, the projects will be further evaluated to determine eligibility for the Market Development track

b) **Market Development (MD) track**: Co-funding support of up to 70% of qualifying costs, capped at $250,000.
The key activities could include but not limited to:
- Develop the solution based on the established blueprint and pilot implementation plan.
- Deploy and test solution on test-bedding sites
- Establish system performance monitoring and to review and validate test results.
- Develop the solution to be market ready

For option (a), please fill up the ASD section. (Optional) MD section can be filled up as well if indicative information is available.
For option (b), please fill up the MD section only.

a. **What are the key activities and deliverables for each phase of the project?**
Please include details such as no. of man-days, cost-breakdown

I. Accelerate Solution Development (ASD) Phase

Phase	Key Activity	Personnel, Role and Man-hours	Resources	Start Date	End Date
ASD 1	User Studies & Feedback Collection	PM & Technical Lead – $6,000/month (full-time) Prepare technical and business proposals, incorporate user feedback, engage in commercial & technical discussions with decision-makers & stakeholders Designer – $3,000/month (full-time) Collect and consolidate user feedback, and translate into UX/UI designs, document customer feedback, contribute to design of the presentation for stakeholder facing collaterals	Various software subscriptions for developing the prototype	January 10, 2023	February 10, 2023
ASD 2	Modify prototype and source relevant data	PM & Technical Lead – $6,000/month (full-time) Prepare technical and business proposals, incorporate user feedback, engage in commercial & technical discussions with decision-makers & stakeholders Designer – $3,000/month (full-time) Collect and consolidate user feedback, and translate into UX/UI designs, document customer feedback, contribute	Various software subscriptions for developing the prototype	February 10, 2023	March 10, 2023

		to design of the presentation for stakeholder facing collaterals Senior Developer – $6,000/month (full-time) Full stack developer to develop the solution based on PM's roadmap and designer's output			
ASD 3	Develop new prototype functionality	PM & Technical Lead – $6,000/month (full-time) Prepare technical and business proposals, incorporate user feedback, engage in commercial & technical discussions with decision-makers & stakeholders Designer – $3,000/month (full-time) Collect and consolidate user feedback, and translate into UX/UI designs, document customer feedback, contribute to design of the presentation for stakeholder facing collaterals Senior Developer – $6,000/month (full-time) Full stack developer to develop the solution based on PM's roadmap and designer's output	Various software subscriptions for developing the prototype	March 10, 2023	April 10, 2023
Overall Project Cost: $ 45,000 ($15,000 per month from January 2023 to April 2023)					
for BEAMP *up to 70% of qualifying costs, capped at $30,000* $ 30,000					

I. Market Development (MD) Phase

Phase	Key Activity	Personnel, Role and Man-hours	Resources	Start Date	End Date
MD 1	Develop data engineering pipeline & Load into the System Propose data science model(s) based on top user priorities	PM – $10,000/month (full-time) Roll out product roadmap based on client feedback, stakeholder engagement, commercial discussions, ensure project is managed on time, schedule, and budget, prepare relevant reporting materials for decision makers, stakeholders, and project teams Engineering Manager - $10,000/month (full-time) Manage engineering tasks, QA, and cybersecurity certifications, as well as integration between	Various software subscriptions for developing the prototype	April 10, 2023	August 10, 2023

		client systems and our product, design and implement ETL data engineering pipelines, deploy data science models Senior Developer – $10,000/month (full-time) Full stack developer to develop the solution based on PM’s roadmap and designer’s output, implement data science models into the platform, resolve QA issues, cybersecurity vulnerabilities, and improvements post-deployment Designer – $3,000/month (full-time) Collect and consolidate user feedback, and translate into UX/UI designs for customer feedback, contribute to design presentation of stakeholder facing collaterals			
MD 2	Deploy data science models, enhance data engineering pipeline, connect newly sourced datasets to augment customers	PM – $10,000/month (full-time) Continue roadmap roll-out on client feedback, stakeholder engagement, commercial discussions Engineering Manager - $10,000/month (full-time) Manage engineering tasks, QA, and cybersecurity certifications, as well as integration between client systems and our product Senior Developer – $7,000/month (full-time) Full stack developer to develop the solution based on PM’s roadmap and designer’s output Designer – $3,000/month (full-time) Collect and consolidate user feedback, and translate into UX/UI designs for customer feedback, contribute to design presentation of stakeholder facing collaterals	Various software subscriptions for developing the prototype	August 10, 2023	December 10, 2023
MD 3	Deploy entire pipeline, add reporting and distribution capabilities,	PM – $10,000/month (full-time) Finalize product roadmap roadmap and prepare final deliverables for commercialization with the	Various software subscriptions for developing the prototype	December 10, 2023	April 10, 2024

			and other stakeholders, incorporate feedback into product discussions with the PM	
To be hired	Senior Developer	Design system architecture, develop software	Lead the design of system architectures	MD 1 -3
To be hired	Engineering Manager	Design system architecture, develop software	Lead the execution of tasks assigned by CEO for technical execution, ensure compliance, quality assurance, bugs, and improvements are corrected	MD 1 -3

Section D: TRACK-SPECIFIC QUESTIONS

Please fill in the sub-section based on the track that you are applying for.

I. For Accelerate Solution Development track

What are the hypotheses that are important to be tested and validated before committing more resources into the project?
Please list down the hypotheses that can be reviewed at the end of Accelerated Solution Development phase to determine the readiness for Market Development phase. Tested hypotheses would produce validated learnings or controlled risks

Priority	Your hypothesis	How might you validate the hypothesis?
1	There is a viable commercial opportunity to bridge enterprise-type data with site-level technical data for the construction sector – and unlock key analytics insights that can drive FP&A function beyond conventional accounting metrics and with greater frequency of reporting (e.g., daily or weekly).	Ask the challenge statement owners whether they currently have an ERP system (either in-house or external vendor). Ask whether they would find it useful to access insights based on this data more frequently. Ask if they would like some type of automated correlation analysis with the day-to-day technical activities from the site.
2	Emission calculations for vehicles and equipment can be economically done using video-based analytics combined with other data sources such as equipment/vehicle log whiles.	Ask the user whether better oversight of billing for energy and fuel can be an avenue to efficient carbon emission measurements. Ask whether supplementing videos with other vehicle or equipment data can be used to aid other key activities for the construction teams.
3	Other companies in the construction sector can benefit from a similar solution (i.e., it's scalable)	Ask the user whether they have a similar set-up and available data at different construction sites. Ask similar companies whether they have a similar need to estimate emissions and resources to allocate towards resolving it.

II. For Market Development track

a. **Please explain the business model.**
Describe the revenue model, key partners and other key aspects of your business that supports the viability of the product / service

Our business is based on a subscription-based SaaS model consisting of different priced tiers. The first tier is a free open source version which can be downloaded on the company's GitHub page. We have a priced enterprise offering that assigns a data expert to support all of the technical activities to empower the available analytics models.

b. **How big is the market demand for the product / service?**

We estimate that a typical construction firm or REIT spends several million dollars per year on ERP software alone. The costs can be even higher for Built-it-Yourself type solutions.

c. Please explain the business impact of the project.

	Now	After Project		
	Last FY	Year 1	Year 2	Year 3
Revenue (SGD)	$0	$1M	$3M	$10M
Net Operating Profit/Loss Before Tax (SGD)	$0	~$400k	~1.2M	~$3M
Remuneration *(i.e. expenses for manpower salaries)*	$0	$600k	$6M	$6M
Depreciation	$0	TBD	TBD	TBD
No. of PMET jobs	1	5	20	50

Section E: CLARIFICATION / REQUEST TO CHALLENGE STATEMENT OWNERS

What are the resources required from the challenge statement owners for a successful pilot?

We would need the relevant datasets for carbon emission estimates: fuel invoices, equipment usage and logs, site footage (showing the vehicle and equipment activity). These can be a historical file or an API connection for real-time analysis and reporting.

Further to the datasets, we would need access to consultations with relevant stakeholders within the organization. This includes both site-level personnel and senior supervisors tasked with oversight.

Do you have any questions for the Challenge Statement Owners?

Q: Can we use additional data to video footage (to supplement its accuracy)?

Q: Are there other analytics solutions that are relevant in addition to the carbon emission estimates?

Q: How frequently does the data need to be collected, calculated, and insights reported?

Upload this completed proposal to the form on beamp.sg,

or ***Mon, , 2359hrs (UTC+8)*** *<submission deadline>*

Please also feel free to submit your pitch decks or additional materials that will aid your submission through the form on beamp.sg.